Date : 18 - 03 - 2026

SQL Execution Plan and Index Optimization

For this task, I used a **sales** database which includes three different datasets for **customers, products, and sales data.**
The main objective of this task was to check how **queries run and improve their speed using indexes.**

I created the **SALES** database and created different tables for customers, orders and sales . Then, I checked the tables using **DESCRIBE** and looked at some sample data using **SELECT** to understand the data.

Then , I wrote different queries like total sales per customer, total sales per product, top selling product, and top customers by sales.

Then, I used **EXPLAIN** before creating indexes to see how the queries were running.

Next, I created indexes on columns like **customer_key** and **product_key.** I also created combined indexes like (customer_key, sales) and (product_key, sales) to make queries faster.

After creating indexes, I used **EXPLAIN** again to compare the results. I observed that using indexes , query execution improved . Hence , performance improved.